AngioLens · *a virtual angiogram*

Coronary artery disease and per-vessel stenosis prediction, exactly explained and mapped onto a 3D heart · **Random Forest Rangers** · Multimodal AI Hackathon 2026 · Track A: Cardiovascular Risk Visualization & Prediction

Problem. Invasive angiography tells a cardiologist *where* coronary disease is. Routine clinical data (history, ECG, labs, echo) arrives earlier and costs far less, but statistical risk scores turn it into a single opaque number. **What we built.** AngioLens predicts overall CAD and ≥50 % stenosis of the LAD, LCX and RCA from 55 routine features of the Z-Alizadeh Sani cohort (n = 303). It explains every prediction *exactly* with SHAP and renders it as a "virtual angiogram": a real BodyParts3D heart whose arteries carry animated contrast flow coloured by risk, and whose myocardium shows the territory each artery supplies. **Key ideas:** (1) a *hierarchical* model P(vessel) = P(CAD)·P(vessel | CAD) that is anatomically coherent by construction; (2) an affine log-odds ensemble that keeps SHAP exactly additive after calibration; (3) conformal prediction that says "indeterminate" instead of guessing; (4) honest out-of-fold replay of all 303 patients; (5) an LLM copilot whose every number is machine-checked against the model output. **Results** (10×5 repeated nested CV): ROC-AUC CAD 0.927 (95 % CI 0.89–0.96), LAD 0.842, LCX 0.737, RCA 0.712; Brier 0.094 for CAD; 90 % conformal coverage held on every target; 60 fps on integrated graphics; 41 ms per explained prediction.

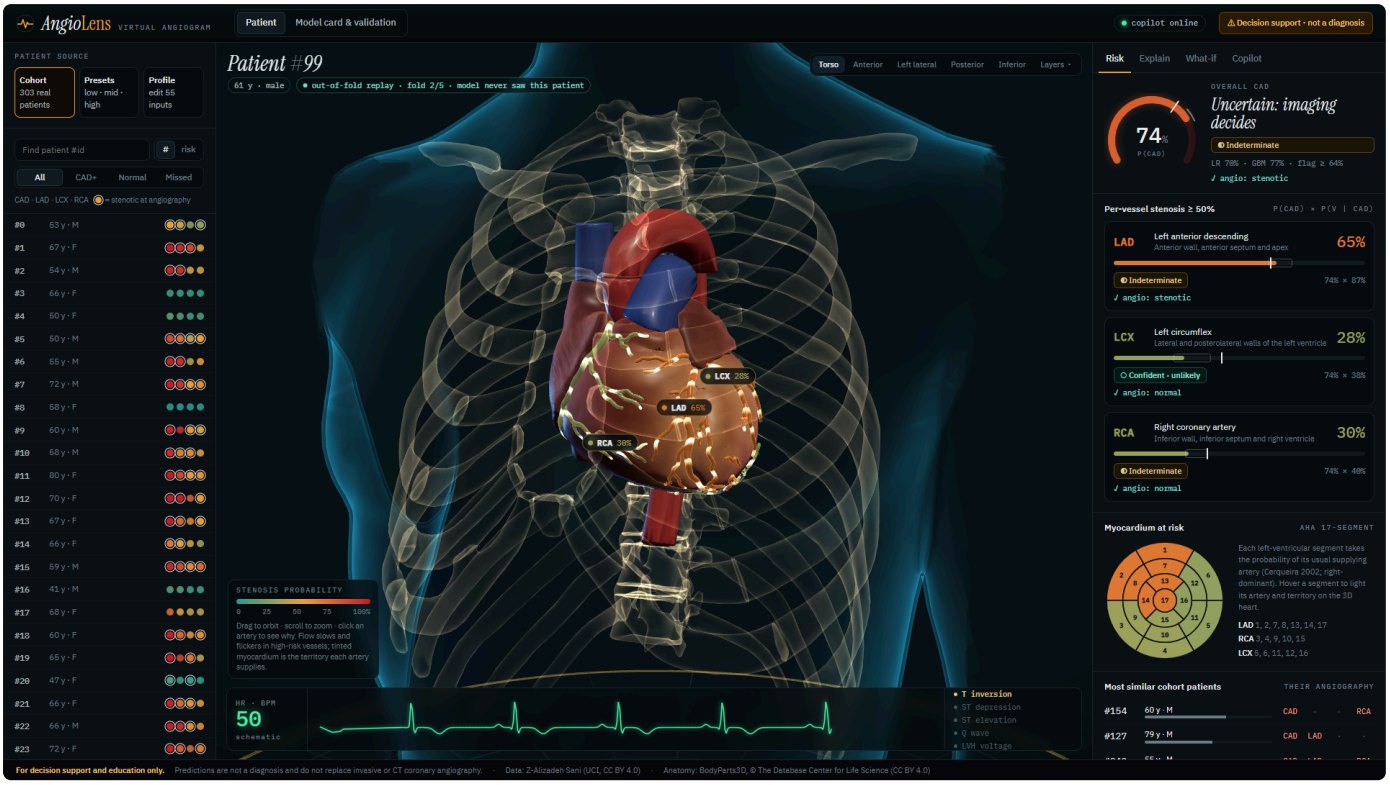


Figure 1. The AngioLens dashboard on cohort patient #99 (single-vessel LAD disease), replayed by the cross-validation model that never saw this patient. Left: cohort browser (out-of-fold risk, angiography truth rings). Centre: BodyParts3D heart inside an x-ray torso; arteries coloured by stenosis probability, myocardium tinted by territory at risk, ECG strip at the recorded heart rate. Right: CAD gauge with conformal status, per-vessel cards with ground truth, AHA 17-segment map.

1 What the system does

- **Predict** overall CAD and LAD / LCX / RCA stenosis with calibrated probabilities, an operating flag, a model range and a 90 % conformal set.
- **Map** each probability onto its artery in 3D (colour, contrast-flow speed, glow) and onto the myocardium it perfuses (3D territory tint + AHA bull's-eye). Rotate, zoom, pick an artery or any point on the heart muscle to inspect it.
- **Explain** with exact SHAP waterfalls in two stages ("is there CAD?" and "where is it?"), clinical-domain totals, and a physiological breakdown of every measurement against its reference range.
- **Probe** with counterfactual what-ifs and per-patient sensitivity curves; compare with the five most similar real patients and their angiograms.
- **Communicate** via a grounded copilot (clinician / patient voice, English / Hindi, voice questions) and a one-page PDF report.

2 Data & preprocessing

Dataset. Extension of Z-Alizadeh Sani (UCI #411, CC BY 4.0): 303 patients who underwent angiography, 55 raw clinical columns plus targets `LAD`, `LCX`, `RCA` and `Cath`. Prevalence: CAD 71%, LAD 58%, LCX 39%, RCA 38%. Diseased vessels per patient: 0 → 86, 1 → 87, 2 → 67, 3 → 63.

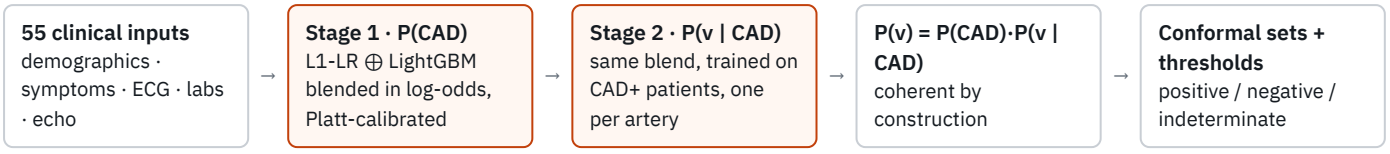
Leakage guard. The four target columns are excluded from the feature registry and the build fails if any reaches the model (asserted in code and in the test suite).

Audit findings. No missing values. `Exertional CP` is constant (all "N") and dropped. One record has `Cath = Normal` although a vessel is stenotic; it is kept unchanged and reported, not silently fixed.

Encoding. A single *feature registry* (`angiolens/schema.py`) declares each of the 55 model inputs once: raw column, clinical domain (demographic, symptom/exam, ECG, lab, echo), unit, valid range, adult reference range and whether it is modifiable. Y/N → 0/1; sex → binary; bundle-branch block → LBBB/RBBB indicators; valvular disease → ordinal 0–3. Training, API validation, the web form and the physiological breakdown are all generated from this registry, so a new clinical feature is a one-line change.

Transforms (fitted inside every training fold only): log1p on right-skewed labs (FBS, creatinine, TG, BUN, ESR, WBC, platelets), then standardisation for the linear branch. Trees use raw values.

3 Model architecture



Hierarchical targets. CAD is defined as ≥50 % narrowing in *any* vessel, so we factorise P(vessel) = P(CAD)·P(vessel | CAD). Stage 2 learns only the localisation question among diseased patients. A vessel probability can never exceed the CAD probability. Independent per-vessel models violate this in 21% of out-of-fold predictions, which would show a heart with a "70 % LAD" next to a "40 % CAD".

Exactly explainable calibrated ensemble. Each stage combines a sparse L1 logistic regression and a shallow LightGBM (500 trees, 8 leaves, strong subsampling) in log-odds space and then applies Platt scaling:

$$z(x) = a \cdot \left(w \cdot z_{LR}(x) + (1 - w) \cdot z_{GBM}(x) \right) + b \quad \Rightarrow \quad \varphi_j(x) = a \cdot \left(w \cdot \varphi^{LR}_j(x) + (1 - w) \cdot \varphi^{TreeSHAP}_j(x) \right)$$

Every step is affine in log-odds, so SHAP values remain **exactly additive** (base + Σφ = calibrated logit to 10^{-6}, unit-tested) while the model keeps the accuracy of the blend. $C \in \{0.05, 0.1, 0.2, 0.4\}$, $w \in \{0, .25, .5, .75, 1\}$ and (a, b) are chosen by inner 5-fold CV on log-loss. Deployed choices: `CAD: C=0.4, w=0.5 · LAD|CAD: C=0.1, w=1.0 · LCX|CAD: C=0.1, w=0.75 · RCA|CAD: C=0.1, w=1.0`.

Uncertainty. Mondrian (class-conditional) conformal prediction at α = 0.10 turns each probability into a prediction set. "Both outcomes" is shown as *indeterminate*. The calibrated LR-vs-GBM spread is shown as a model range. Operating thresholds maximise Youden's J on inner out-of-fold predictions (CAD 0.72 · LAD 0.66 · LCX 0.38 · RCA 0.35).

4 Validation methodology & predictive performance

10×5 **repeated, stratified, nested** cross-validation (strata = CAD × number of diseased vessels). Hyper-parameters, calibration, thresholds and conformal quantiles are fitted inside each outer training split, so no outer-test patient ever influences a modelling decision. 95 % CIs combine patient bootstrap with repeat-to-repeat CV variation (1000 draws). The full run is deterministic and takes 265 s on a laptop CPU.

TARGET	PREV.	ROC-AUC (95 % CI)	PR-AUC	BRIER	ECE	PREC.*	RECALL / SPEC*	ACC*	F1*	MCC*	ACC · F1 @0.5
Overall CAD	71%	**0.927** 0.89–0.96	0.969	0.094	0.040	94%	86% / 86%	86%	0.895	0.679	88% · 0.916
LAD stenosis	58%	**0.842** 0.80–0.88	0.869	0.155	0.053	80%	83% / 70%	78%	0.814	0.539	78% · 0.828
LCX stenosis	39%	**0.737** 0.68–0.79	0.588	0.199	0.040	55%	79% / 58%	66%	0.648	0.367	67% · 0.578
RCA stenosis	38%	**0.712** 0.64–0.78	0.545	0.203	0.045	51%	75% / 57%	64%	0.607	0.312	65% · 0.503

* at the operating threshold selected by Youden's J in the inner CV (honest: chosen without the outer test fold).

Competing models, identical outer splits: mean ROC-AUC (Brier)

MODEL	OVERALL CAD	LAD	LCX	RCA	COHERENT OUTPUT†
AngioLens (hierarchical blend)	**0.927** (0.094)	0.842 (0.155)	0.737 (0.199)	0.712 (0.203)	100.0%
Direct LR+GBM blend	**0.927** (0.094)	0.841 (0.157)	0.734 (0.204)	0.719 (0.205)	79.5%
L1 logistic regression	0.926 (0.096)	0.831 (0.163)	0.708 (0.212)	**0.722** (0.204)	79.2%
LightGBM	0.918 (0.098)	0.843 (0.155)	0.734 (0.204)	0.708 (0.207)	78.7%
Random forest	0.920 (0.114)	**0.851** (0.164)	**0.739** (0.205)	0.703 (0.208)	96.7%

† share of predictions with every P(vessel) ≤ P(CAD). Differences between models are within one standard deviation for most targets. AngioLens matches the best discrimination while giving the best or tied-best Brier score, exact explanations and 100 % coherence.

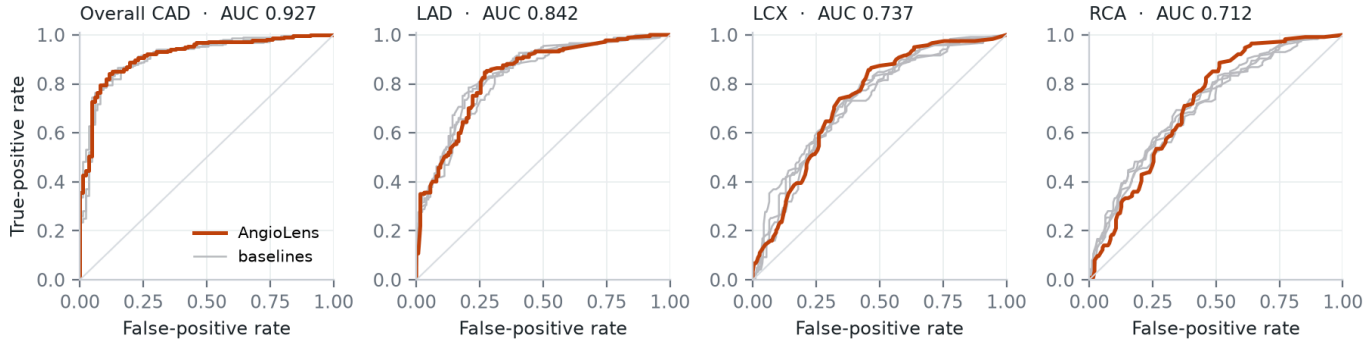


Figure 2. Out-of-fold ROC curves (repeat 1). AngioLens in orange, the four baselines in gray.

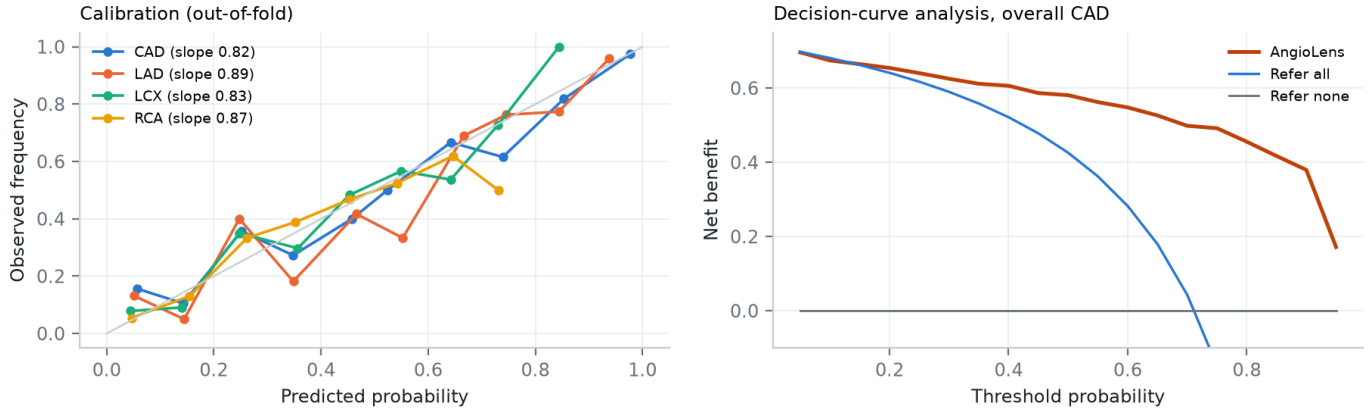


Figure 3. Left: reliability of out-of-fold probabilities (slope 1 = perfect). Right: decision-curve analysis for CAD. AngioLens gives higher net benefit than "refer everyone" across referral thresholds of roughly 10–90 %.

Conformal prediction: does "90 % confident" hold out-of-sample?

TARGET	COVERAGE	INDETERMINATE	ACC. WHEN DECIDED	CAL. SLOPE
CAD	90.6%	9%	89.7%	0.823
LAD	89.3%	31%	84.5%	0.887
LCX	89.2%	50%	78.4%	0.831
RCA	88.6%	51%	76.8%	0.870

Reading the numbers honestly

Overall CAD is predicted well (AUC 0.927). When the model commits, it is right 89.7% of the time while abstaining on 9% of patients. **Localising** the lesion from routine data is much harder: LAD is good (AUC 0.842), LCX/RCA are moderate. The system says so, flagging 50% / 51% of LCX/RCA predictions as indeterminate rather than painting a confident heart. Published per-artery accuracies of 83–86 % on this dataset (Alizadehsani et al., 2020) were not obtained under nested validation, so they are not directly comparable.

5 Clinical interpretability

- **Two-stage exact SHAP.** "Why CAD?" (stage 1) and "why this artery, given CAD?" (stage 2) waterfalls on a log-odds axis labelled in probability. Contributions sum exactly to the displayed probability. Clicking an artery in 3D switches the explanation to its stage.
- **Physiological breakdown.** Every measurement with value, unit, reference-range bar, status (high / low / normal) and its signed contribution; totals per clinical domain (demographics, symptoms, ECG, labs, echo).
- **Case-based evidence.** The five nearest cohort patients (distance weighted by global SHAP importance) with their angiography results.

- **Counterfactual probes.** "If chest pain were atypical…" and sensitivity curves. These revealed *confounding by treatment*: setting LDL to 70 mg/dL *raises* modelled CAD risk for 64 % of patients, because LDL is no higher in CAD patients (104 vs 106 mg/dL; many are on statins). The UI flags such results as counter-intuitive associations, not treatment effects.

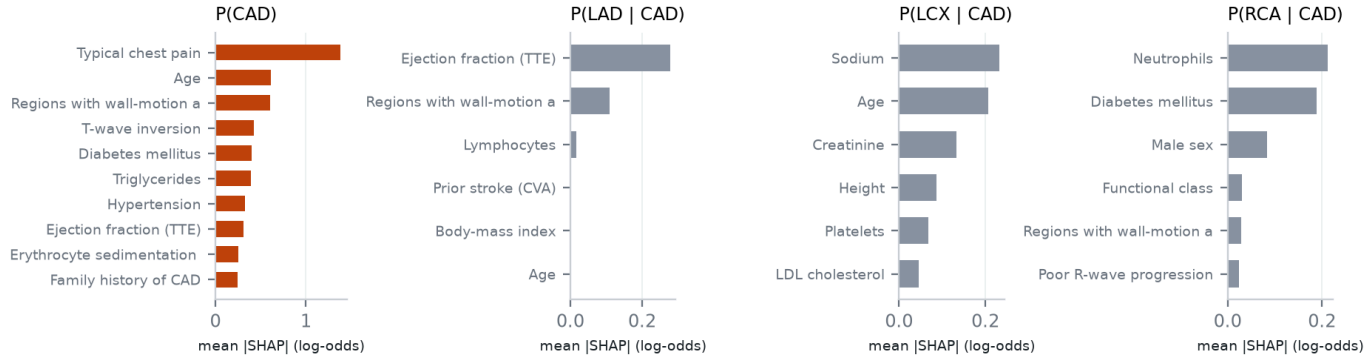


Figure 4. Global mean |SHAP| of the deployed model. Stage 1 is dominated by typical chest pain, age, wall-motion abnormality and ECG changes, which is clinically plausible. Stage-2 localisers are weak and partly lab-based, consistent with the moderate LCX/RCA performance.

6 3D anatomical pipeline

Source. BodyParts3D 4.0 (DBCLS, CC BY 4.0), selected by FMA id: LAD tree (anterior interventricular branch + diagonals, conus and right-anterior branches, 18 meshes), circumflex (6), RCA tree (trunk, marginal, posterior descending, ventricular and conus branches, 29), left main, four chambers, cardiac veins, aorta, pulmonary trunk, SVC, lungs, rib cage, spine, clavicles and the torso skin.

Epicardial correction. BodyParts3D models ventricular *cavities*, so arteries floated 9 mm (LV) and 4 mm (RV) above the surface. We measured these gaps, subdivided and Taubin-smoothed the shells, and offset them along normals (7.5 / 3 mm). Arteries now sit 1–2 mm from the epicardium.

Baked attributes (`scripts/build_anatomy.py`): for every artery vertex its distance downstream of the coronary ostium (found as the vertex nearest the aortic root), which drives the flow shader. For every ventricular vertex, soft perfusion-territory weights for LAD / LCX / RCA (softmin of distance to each artery tree, $\tau = 7$ mm, proximity falloff), which drive the territory tint and region picking. Output: one GLB (4.9 MB, 194 k triangles) with named nodes, so model outputs map 1:1 to `LAD`, `LCX`, `RCA`.

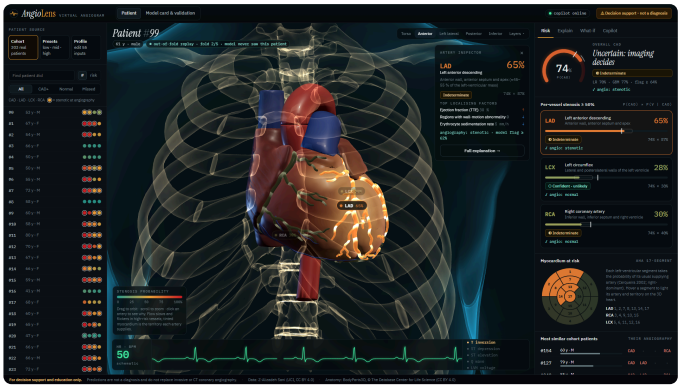


Figure 5. Anterior view, LAD selected: the artery is emphasised, other vessels and bull's-eye segments dim, and the inspector lists its probability, conformal status, localising factors and angiography truth.

Rendering (React Three Fiber). Custom GLSL: arteries show contrast boluses travelling from ostium to apex that *slow down and flicker* as stenosis probability rises, plus an additive halo shell instead of a bloom pass. Myocardium uses per-vertex territory colouring. The torso and bones use a fresnel x-ray shader with a scan line. The heart beats at the patient's recorded pulse, in sync with a schematic ECG whose morphology reflects the recorded ST / T / Q / BBB / LVH findings. No post-processing and no float render targets: **60 fps at 1080p on Intel Iris Xe integrated graphics**, and it also runs on software WebGL.

7 System integration & extensibility

A FastAPI backend serves the pre-built React app and a typed REST API: `/api/predict` (prediction + exact SHAP for all four stages + similar patients, **41 ms** median), `/api/cohort/{id}` (honest out-of-fold replay), `/api/sensitivity` (11 ms), `/api/encode` (raw UCI rows / CSV / JSON import), `/api/copilot`, `/api/transcribe`, `/api/report` (PDF) and `/api/model-card`. Every slider move re-predicts (debounced 90 ms, in-flight requests aborted), so the arteries, territory tint, gauge, waterfall and ECG update in real time. Extending the system is local: a new **clinical feature** is one registry entry plus data; a new **model** is a `Stage` with `decision()` and `shap()`; a new **anatomical structure** is one FMA id in the build script and one node name in the viewer.

Grounded copilot. The LLM (Groq `gpt-oss-120b`) never predicts. It rephrases a deterministic fact sheet derived from the model output. Every number in its answer (Devanagari digits included) is checked against that sheet; on failure it is asked to rewrite once, then replaced by a template. Out-of-scope questions (dosing, targets) are declined. Without an API key everything works offline.

8 Usage

`pip install -r requirements.txt` → `python run.py` → open `http://127.0.0.1:8000`. The trained model, built anatomy and built web app ship in the repository. To reproduce from scratch: `python scripts/train.py` (nested CV, ~5 min), `python scripts/build_anatomy.py` (downloads BodyParts3D), `cd web && npm ci && npm run build`; tests: `pytest` (20 tests). Optional `.env` with `GROQ_API_KEY` enables the LLM and voice. Docker: `docker build -t angiolens . && docker run -p 8000:8000 angiolens`.

9 Limitations & safety

Decision-support and educational prototype, not a medical device. The UI shows a persistent disclaimer bar, an acknowledgement dialog, conformal "indeterminate" states and disclaimers in every PDF and copilot answer.

- Single-centre, 303-patient cohort; no external validation. Performance will differ in other populations and referral settings.
- Vessel localisation is moderate (LCX/RCA AUC ≈ 0.71–0.74). The dataset contains no lesion location, severity grade or troponin.
- The 3D heart is a **generic** anatomy (right-dominant). Territory mapping follows standard AHA assignments, not this patient's coronary tree.
- SHAP and what-ifs describe model associations, which are confounded by treatment. They are not causal effects or treatment advice.
- The ECG strip is schematic. The copilot can only rephrase; it inherits every limitation above.

10 References

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